

Question	Answer	Marks
4(a)	acceleration proportional to displacement	B1
	acceleration <u>directed</u> towards fixed point or displacement and acceleration in opposite <u>directions</u>	B1
4(b)(i)	1. amplitude decreases gradually so light damping or oscillations continue so light damping	B1
	2. loss of energy	B1
	due to friction in wheels or due to friction between wheels and surface (during slipping) or due to air resistance (on trolley)	B1
4(b)(ii)1.	$\omega^2 = 2k/m$	C1
	$= (2 \times 230) / 0.950$	C1
	$\omega = 22 \text{ rad s}^{-1}$	A1
4(b)(ii)2.	$T = 2\pi / \omega$	C1
	$T = (2\pi / 22) = 0.286 \text{ s}$ time $= 1.5T$ $= 0.43 \text{ s}$	A1

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$f(\omega)$	range of frequencies (of signal)	D1